

Study on runoff forecasting and error correction driven by atmosphere–ocean–land dataset

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ABSTRACT

Accurate runoff forecasting results can not only provide an important basis for flood control scheduling, but also provide scientific support for water resources optimization, which promotes the maximization of the overall benefits of the basin. To further explore the inherent mechanisms of the atmosphere–ocean–land factors driving runoff changes, this study proposes the factors dimension reduction and interpretation framework based on Pearson, eXtreme Gradient Boosting and SHapley Additive exPlanations (P-XGBoost-SHAP). Based on this, the Gaussian Process Regression (GPR), Long Short-Term Memory neural network (LSTM) and Support Vector Machine (SVM) models are used to construct the atmospheric–ocean–land data-driven runoff prediction model. Meanwhile, for the runoff prediction residuals, this paper proposes an error multi-step correction framework based on ensemble empirical mode decomposition and autoregressive model (EEMD-AR). The case study of Lianghekou hydrological station shows that the factor dimension reduction and interpretation framework can greatly reduce the input dimension of the model, and explain the factors globally and locally by using SHAP Value. Compared with the traditional Random Forest (RF) dimension reduction method, it shows higher prediction accuracy. The Nash–Sutcliffe efficiency coefficient (NSE) can be increased to about 0.93, which is 4.91 % and 1.97 % higher than the series–parallel coupling (AR-Parallel) and empirical mode decomposition–autoregressive (EMD-AR) correction methods, respectively. The accuracy of the runoff forecasting prediction is improved while reducing the input dimension of the model.

1. Introduction

Accurate and reliable medium and long-term runoff forecasting can provide important reference for flood control and drought relief in the basin and scientific operation of reservoirs, and give full play to the safety and economic benefits of reservoir groups (Feng et al., 2020; Xu et al., 2024; Xu et al., 2022). However, the physical mechanism of the formation of medium and long-term runoff process is complex, involving many predictors, and the forecast period is long, and the time-varying and non-stationary properties of runoff are prominent, which makes it difficult to excavate the regular characteristics of runoff formation (J. Guo et al., 2023; Karimi et al., 2016). Therefore, how to integrate multi-source heterogeneous information and improve the accuracy and reliability of medium and long-term runoff forecasting is still a difficult problem for many scholars. At present, the common runoff forecasting models include process-driven model and data-driven model. The

process-driven model explores the internal physical mechanism and law of rainfall runoff, and has been widely used in the field of runoff forecasting (Bhasme & Bhatia, 2024; Lin et al., 2023; Yuan et al., 2018). However, with the development of artificial intelligence technology, data-driven runoff forecasting models have become a hot topic for many scholars (Moosavi et al., 2022; Wang et al., 2023; Zhang & Yan, 2023).

The runoff process of the basin is the result of the interaction of multiple factors, which involves many factors such as runoff, temperature, precipitation, atmospheric circulation, etc. If all of them are taken into account in the modeling process, it will cause data redundancy and complex models. It is limited to reveal the objective law and change mechanism of runoff formation. In addition, many scholars have found that precipitation and runoff in the basin are closely related to tele-related climatic factors such as atmospheric circulation situation, monsoon index and sea surface temperature anomalies in different regions (Lai et al., 2022; Maddu et al., 2022) hydrological, meteorological,

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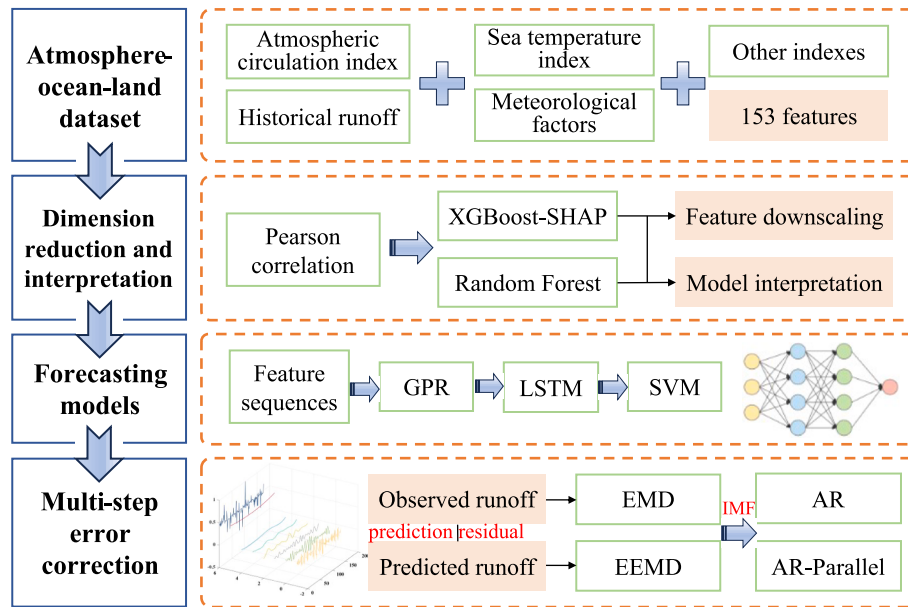


Fig. 1. Research work flow chart.

and climatic variables interact and exert mutual influences (Fang et al., 2023; Ruiz et al., 2007). However, the medium and long-term runoff forecasting methods often use historical runoff or meteorological factor data as model driving factors, ignoring the interactive influence of global teleconnection factors, unable to deeply analyze the physical mechanism of runoff cycle, and have limited effect on improving the accuracy of runoff forecasting (Yang et al., 2023; Yao et al., 2023). eXtreme Gradient Boosting (XGBoost) is an efficient gradient boosting decision tree algorithm. It has been improved on the basis of the original Gradient Boosting Decision Tree (GBDT), which greatly improves the model effect. Its core is to adopt the idea of Boosting to form a powerful ensemble model by integrating multiple weak learners (X. Guo et al., 2023; Sun et al., 2024). In XGBoost, the importance of features can be evaluated by calculating the segmentation contribution of features in the tree model.

The construction of forecasting model is a key step in medium and long-term runoff forecasting, and it is also the core part of medium and long-term runoff forecasting. The advantages and disadvantages and applicability of forecasting model directly determine the prediction accuracy and reliability. In recent years, with the advancement of computer technology and intelligent algorithms, a modern hydrological forecasting system represented by artificial intelligence algorithms is gradually being established (Mo et al., 2023). Deep learning has strong nonlinear ability and generalization characteristics, which can effectively predict based on the knowledge learned from the training set. It plays a significant role in excavating the internal mechanism of rainfall-runoff and analyzing the formation law of runoff, and has been widely used in the field of hydrological forecasting (Tan et al., 2018; Zhang et al., 2021). At the same time, due to the limitations of a single model, it cannot meet the requirements of runoff forecasting in different basins. The combined forecasting method can couple the advantages of different models and is also one of the research hotspots in practical applications (Chen et al., 2015). However, the internal mechanism and decision-making process of deep learning are difficult to understand intuitively, which makes it often regarded as a black box model. This limits the application of deep learning technology in hydrological simulation. For this reason, the Shapley Additive exPlanations (SHAP) interpretable framework provides a new way to explain deep learning. SHAP can explain the 'black box model' from both global and local levels. Its wide application has deepened the depth and level of understanding of complex hydrological systems (Madhushani et al., 2024;

Wang et al., 2022; Wang & Peng, 2024).

The runoff process obtained by the hydrological forecasting model is essentially an average state of flow, and its forecast error is inevitable. Therefore, it is necessary to correct the forecast error in real time (Han & Morrison, 2022). Error correction is the supplementary research in hydrological forecasting, which can improve the accuracy and reliability of forecasting and enhance the applicability and generalization ability of the model. At present, the common error correction methods include error autoregressive, least square method, series-parallel coupling and Kalman filter (Chen et al., 2015; Song & Wang, 2022; Wu et al., 2012). The above error correction method is to reduce the error and improve the accuracy by preprocessing the prediction residual sequence and then predicting. Mode decomposition can deal with non-stationary nonlinear signals and identify time-varying characteristics in the error sequence. It has incomparable advantages in the preprocessing of prediction residuals (Ali et al., 2020; Ius et al., 2024; Rezaie-Balf et al., 2019). This can better explore the potential information of the forecast residual sequence to a certain extent, weaken the redundant information of the original sequence, and thus improve the forecast accuracy (Chen et al., 2021).

The key contribution that has been followed for this research work is detailed below,

- Firstly, the historical runoff, meteorological factors and circulation indexes are used as the driving factors of the runoff model, and the response degree of runoff and characteristic factors is preliminarily analyzed in Section 4.1.
- Then, Section 4.2 proposes the feature dimension reduction and factor interpretation framework. While determining the optimal model input dimension, the global interpretation and interaction analysis of the model input variables are performed.
- Subsequently, in Section 4.3, a data-driven medium-and long-term runoff model is constructed to achieve high-precision simulation and prediction of runoff processes.
- Finally, Section 4.4 proposes the multi-step prediction error 'decomposition-prediction-integration' correction framework to improve the accuracy and level of runoff forecasting.

The remaining sections are organized as follows. Section 2 introduces the factor dimensionality reduction and interpretation framework, runoff forecasting models and error correction framework in

Table 1
Pearson correlation coefficient and correlation degree.

Pearson	Degree of correlation
$ r > 0.8$	Strongly related
$0.6 < r \leq 0.8$	Strong correlation
$0.4 < r \leq 0.6$	Moderate
$0.2 < r \leq 0.4$	Weak correlation
$ r \leq 0.2$	Very weak correlation

detail. In section 3, the application case and comparison model are given, and the analysis and discussion are carried out in section 4. Finally, the conclusion is given in section 5. The overall process of this study is shown in Fig. 1.

2. Methodology

2.1. Dimension reduction and interpretation framework

2.1.1. Pearson correlation coefficient

Pearson correlation coefficient is used to describe the closeness of the linear relationship between two variables X and Y , which is generally expressed by r_{xy} . It is widely used in the correlation screening analysis of hydrological and meteorological runoff predictors (Akbarian et al., 2023). In the correlation analysis of two data sequences $X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n)$, the calculation formula of Pearson correlation coefficient is as follows:

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

where, $\bar{x} = \sum_{i=1}^n x_i / n$ is the mean value of the data sequence X ; $\bar{y} = \sum_{i=1}^n y_i / n$ is the mean value of the data sequence Y ; n is the length of data sequence.

The range of Pearson correlation coefficient r is $[-1, 1]$. When the $|r|$ value is closer to 1, it indicates that the correlation between variable X and variable Y is stronger. The relationship between the degree of correlation between general variables and the range of Pearson correlation coefficient r is shown in Table 1.

2.1.2. eXtreme Gradient Boosting

eXtreme Gradient Boosting (XGBoost) is an ensemble model based on decision trees and the boosting algorithm, which is introduced by Chen and Guestrin (2016) in their seminal work. The fundamental principle of the boosting algorithm is as follows: it starts by creating a weak learner

based on the original samples, and then generates new weak learners based on the fitting error of the previous one. Each weak learner is constructed with reference to the loss function in each step, and the final strong learner is formed by the weighted aggregation of all weak learners (Dong et al., 2023). The core innovation of the XGBoost algorithm lies in its optimization and enhancement over traditional gradient boosting trees, incorporating regularization and second-order Taylor expansion into the loss function 'obj', which is defined as follows (Bentéjac et al., 2021; Osman & A et al., 2021; Qiu et al., 2022).

$$\text{obj} = \sum_{i=1}^m l(y_i, \hat{y}_i) + \sum_{k=1}^t \Omega(f_k) \quad (2)$$

$$\hat{y}_i = \sum_{k=1}^{t-1} f_k(x_i) + f_t(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i) \quad (3)$$

where, y_i is the actual value; $\hat{y}_i$ is the predicted value; l is the loss function, which is generally mean square error, average relative error, etc; m represents the amount of data of the t -th tree; t is all the trees currently established; f_k is the k -th decision tree; Ω represents the complexity of the model.

Substituting equation (2) into equation (3) and applying a second-order Taylor expansion at point $\hat{y}_i^{(t-1)}$ for the first term on the right-hand side yields.

$$\text{obj} = \sum_{i=1}^m [l(y_i, \hat{y}_i^{(t-1)}) + f_t(x_i)g_i + \frac{1}{2}(f_t(x_i))^2 h_i] + \sum_{k=1}^{t-1} \Omega(f_k) + \Omega(f_t) \quad (4)$$

As all terms involving $l(y_i, \hat{y}_i^{(t-1)})$, $\sum_{k=1}^{t-1} \Omega(f_k)$ in equation (4) are constants, the objective function is transformed into.

$$\text{obj} = \sum_{i=1}^m [f_t(x_i)g_i + \frac{1}{2}(f_t(x_i))^2 h_i] + \Omega(f_t) \quad (5)$$

$$\Omega(f) = \gamma T + \frac{1}{2} \alpha \sum_{j=1}^T |w_j| + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (6)$$

where, g_i and h_i are the first derivative and the second derivative of $l(y_i, \hat{y}_i^{(t-1)})$ at $\hat{y}_i^{(t-1)}$, respectively, which are collectively referred to as the gradient statistics of the sample; T is the number of leaf nodes in the decision tree; w_j is the sample weight on the leaf node; γ, α, λ is the model complexity parameter.

Compared with random forest (RF) to construct independent decision trees, each decision tree in XGBoost algorithm is generated by iterative training of minimizing loss function on the basis of existing

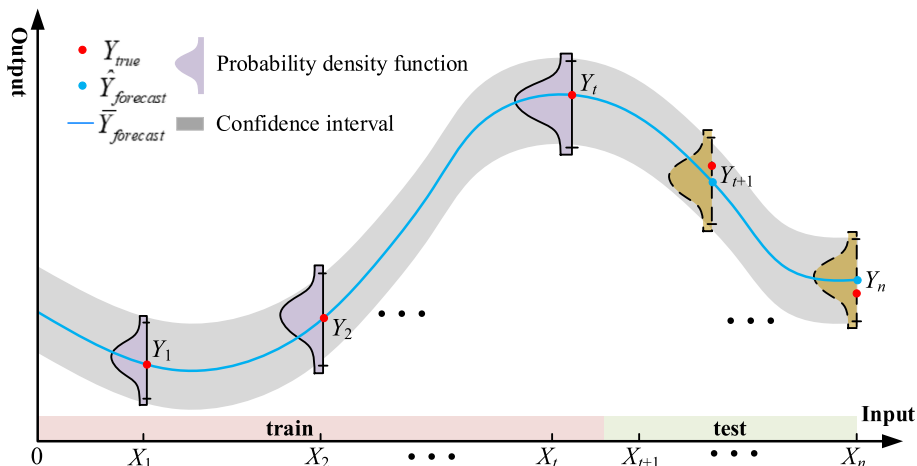


Fig. 2. The generalized diagram of Gaussian process regression.

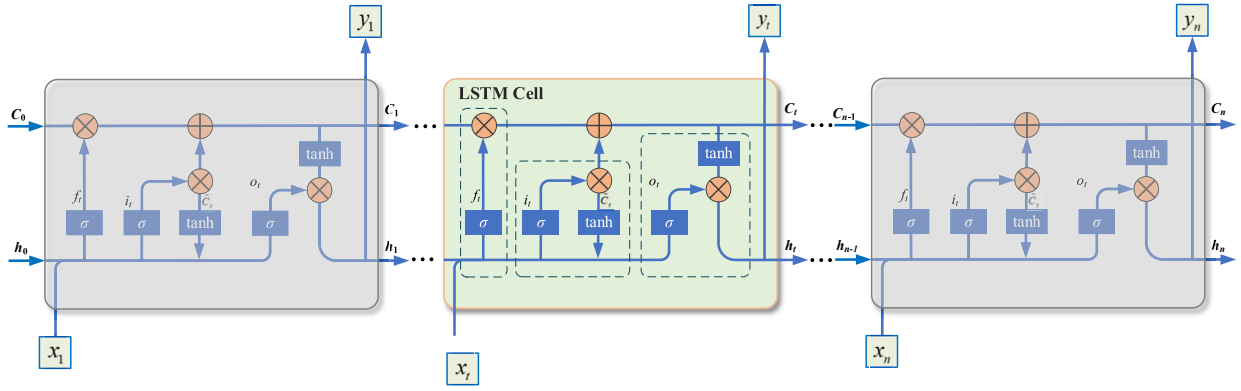


Fig. 3. LSTM neural network structure.

decision trees. In addition, it improves the efficiency of the algorithm through column sampling technology, and introduces regular terms to limit the complexity of the model, which reduces the model over-fitting to a certain extent. In terms of evaluating the importance of factors, XGBoost provides a more refined measure of feature importance, which maintains good computational performance while reducing the complexity of dimensions and models.

2.1.3. Shapley Additive exPlanations

Although XGBoost has excellent regression and classification performance, its internal mechanism and decision-making process are difficult to understand and are considered to be black box models. SHAP plays a vital role in the field of XGBoost regression, which enhances the transparency and interpretability of machine learning methods (Gupta et al., 2024). SHAP is a post-interpretation method based on game theory. SHAP measures the influence of each feature by calculating the marginal contribution of each feature in the model, and then explains the black box model. This marginal contribution is called Shapley Value in SHAP. This is very suitable for explaining the black box model of XGBoost. In addition, SHAP also provides extremely powerful data visualization capabilities, such as interactive interpreters and decision tree visualization. We use Python SHAP package and TreeExplainer function to generate SHAP values, and then get the dependency graph (Bacanan et al., 2024).

2.2. Runoff forecasting models

2.2.1. Gaussian process regression

Gaussian process regression (GPR) is a non-parametric data-driven model for regression analysis of data proposed (Rasmussen, 2003; Williams & Rasmussen, 1995). The model is constructed based on the Bayesian probability framework, which mainly includes regression residuals and Gaussian process priors. Compared with the traditional regression model, GPR can perform both deterministic prediction and uncertainty estimation (Sun et al., 2014). The GPR performs well under small samples, and does not need to train a large amount of data to complete the high-dimensional regression problem (Uddin et al., 2023). The generalized diagram of Gaussian process regression is shown in Fig. 2. Meanwhile, the GPR has good sparse data processing and nonlinear modeling capabilities, which is particularly suitable for the study of medium and long-term runoff prediction. In this study, the Gaussian kernel function is used to capture the complex nonlinear relationship between the characteristic factors and the predicted flow.

2.2.2. Long Short-Term memory

Long short-term memory neural network (LSTM) is a special recurrent neural network mainly composed of input gate, output gate and forgetting gate, as shown in Fig. 3. It has the recursive property of RNN and can effectively use long sequence information, and has good

performance in various fields (J. Guo et al., 2023). At the same time, LSTM can capture and remember the information in the data sequence, and is not easily affected by the problem of gradient disappearance or gradient explosion, which makes them perform well in the fields of natural language processing, speech recognition and time series analysis (Hochreiter & Schmidhuber, 1997). Due to the special memory mechanism and gate structure design of LSTM, it can flexibly adapt to the correlation characteristics contained in learning time series data, so it is widely used in sequence model analysis (Greff et al., 2016).

2.2.3. Support vector Machine

Support vector machine (SVM) was proposed by (Vapnik, 1999). This method combines the VC dimension theory of statistics and the machine learning method of structural risk minimization. It was first used to solve pattern recognition problems. It shows many unique advantages in solving small sample, nonlinear and high-dimensional pattern recognition. It can analyze data, identify patterns, and be used for classification and regression analysis, which also makes it occupy an important position in medium and long-term runoff forecasting (Tang et al., 2019), and has broad development prospects. Its objective function and constraint conditions are calculated as follows:

$$\begin{cases} \min \left[\frac{1}{2} \|w\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \right] \\ \text{s.t.} \begin{cases} y_i - w\Phi(x_i) - b \leq \varepsilon + \xi_i \\ -y_i + w\Phi(x_i) + b \leq \varepsilon + \xi_i^*, i = 1, 2, \dots, l \\ \xi_i \geq 0, \xi_i^* \geq 0 \end{cases} \end{cases} \quad (7)$$

The final regression function is:

$$f(x) = \sum_{i=1}^l (a_i - a_i^*) K(x_i, x) + b \quad (8)$$

where, $\Phi(x)$ is a nonlinear mapping function; $\|w\|^2/2$ is the regularization part; b is the offset of the hyperplane; ξ_i and ξ_i^* are relaxation variables; ε is the estimation accuracy; a_i, a_i^* is the Lagrange multiplier in the quadratic programming; $K(x_i, x_j) = \Phi(x_i)\Phi(x_j)$ is a kernel function that satisfies the Mercer condition. In this study, the radial basis kernel function is selected as the kernel function of SVM.

Due to the strong spatial and temporal variation characteristics of the runoff process, there is a complex mapping relationship with many factors (such as rainfall, evaporation, land use type, etc.). SVM can capture the nonlinear relationship between runoff data and characteristic factors, which plays an important role in solving the nonlinear characteristics and uncertainty factors in actual hydrological problems. At the same time, SVM can effectively deal with high-dimensional data,

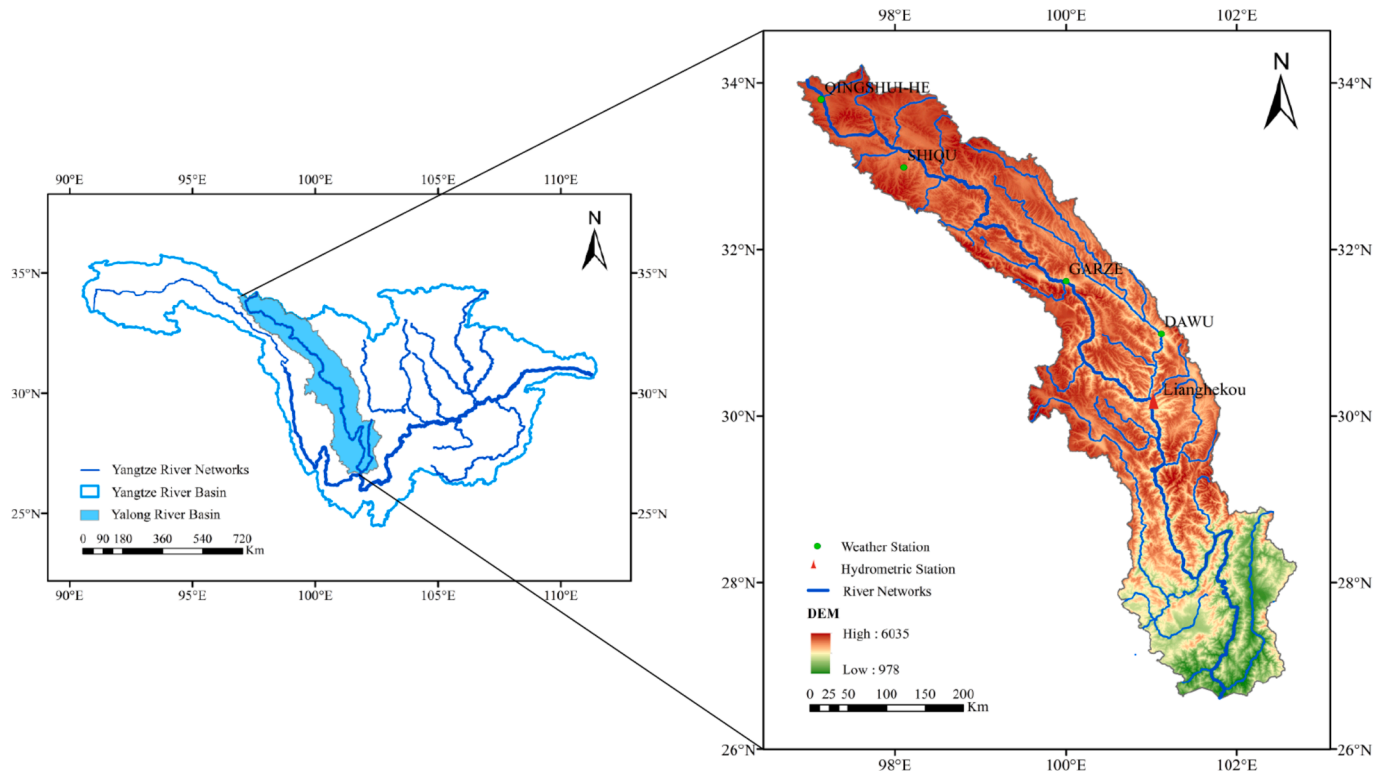


Fig. 4. Study area and site diagram.

has good generalization performance, and can better adapt to the inspection period data.

2.3. Multi-step error correction framework

The hydrological forecasting model is a mathematical method to generalize the physical mechanism of the basin. Therefore, hydrological forecasting inevitably has certain errors, and the physical basis of different hydrological forecasting models is different. The forecasting performance of each model under different regional conditions and different forecasting periods is quite different. It is difficult to further improve the prediction accuracy of a single prediction model. In order to improve the accuracy of runoff forecasting model, this paper proposes a 'decomposition-prediction-reconstruction' error correction framework based on ensemble empirical mode decomposition-autoregressive (EEMD-AR). Firstly, EEMD modal decomposition is carried out based on the runoff forecast residuals of each model. Secondly, AR model prediction is carried out for each modal component, and then the predicted modal components are integrated to obtain the runoff forecast value after error correction. The following discusses the correction prediction method.

(1) Ensemble empirical mode decomposition

In order to study the signal processing method, (Wu & Huang, 2009) proposed the empirical mode decomposition algorithm (EMD). This method is to decompose the waveform or trend of different scales in any signal step by step, generate a series of data sequences with different characteristic scales, and treat each sequence as a characteristic modulus function component. Compared with Fourier transform, it can deal with non-stationary and nonlinear signals, decompose complex signals into residual waves and a series of intrinsic mode functions (IMF). The final decomposition results of EMD are as follows (Jin et al., 2023; Yan et al., 2022):

$$x(t) = \sum_{i=1}^n imf_i(t) + c(t) \quad i = 1, 2, \dots, n \quad (9)$$

where, t is time, n is modal number; $x(t)$ is the original signal; $c(t)$ is the residual component; $imf_i(t)$ is the i -th IMF obtained by EMD.

The IMF obtained by the EMD decomposition has a modal aliasing phenomenon, which will cause signals of different scales or frequencies to be mixed into the same IMF, making it difficult to identify its characteristics. In order to avoid mode mixing, (Wu & Huang, 2009) proposed a new noise-assisted data analysis method, namely ensemble empirical mode decomposition (EEMD). Its core idea is to add Gaussian white noise to the signal for multiple EMD decomposition, and finally define the overall average of the multiple decomposed IMFs as the final IMF, so as to achieve the purpose of eliminating the added white noise, and then effectively suppress the generation of modal aliasing. The decomposition results of EEMD are as follows:

The EMD test is carried out on M times, and the mean values of all IMF components and residual components are obtained as follows:

$$\bar{x}_i(t) = \sum_{m=1}^M x_{i,m}(t) / M \quad i = 1, 2, \dots, n \quad (10)$$

$$\bar{c}_n(t) = \sum_{m=1}^M c_{i,m}(t) / M \quad i = 1, 2, \dots, n \quad (11)$$

where, M is the total number of EMD tests; $\bar{x}_i(t)$ and $\bar{c}_n(t)$ are the n -th IMF component and the remaining component obtained by EEMD decomposition, respectively.

(2) Autoregressive model.

The autoregressive model (AR) is a method of processing time series data in statistics. Its basic idea is to analyze the statistical law of the evolution of time series data and the correlation between the elements at different times (Hasebe et al., 1989). When the error correction is carried out by the AR model, the modal component of the previous predicted runoff residual data is used to infer the modal component of the current runoff residual value in order to achieve the purpose of error correction. The AR model has a simple structure and is easy to implement. It is widely used in hydrological runoff prediction. The model

expression is as follows (Malakoutian et al., 2022):

$$X_t = \varphi_0 + \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + \dots + \varphi_p X_{t-p} + \varepsilon_t \quad (12)$$

where, X_t is time series data; φ_0 is a constant term, and $\varphi_0 \neq 0$; $\varphi_1, \varphi_2, \dots, \varphi_p$ is an autoregressive coefficient, p is an autoregressive order; ε_t is the normal distribution white noise signal with the mean value of 0 and the variance of σ_ε^2 .

In this study, EEMD was selected to decompose the residual of runoff forecast to obtain each modal component (IMF), and then AR was used to re-forecast each IMF. Finally, the predicted modality component IMF is integrated, and then the predicted residual sequence is combined with the original forecast value to obtain the error-corrected runoff forecast value, so as to achieve the purpose of error correction.

3. Case study

3.1. Study object

This study focuses on the Lianghekou hydrological station in the Yalong river basin and discusses and analyzes the proposed model and methods. The Yalong river basin boasts a main stem that extends over 1571 km, featuring a natural drop of 3830 m and covering a watershed area of 136,000 square kilometers, as shown in Fig. 4. The annual average precipitation in the basin ranges from 520 mm to 2470 mm, increasing from north to south. At the river mouth, there is an average annual flow of 1930 cubic meters per second, with an annual runoff of nearly 60 billion cubic meters, constituting 13.3 % of the total water resources in the upper reaches of the Yangtze River. Due to its ample water supply, significant waterfalls, and concentrated flow, the Yalong river basin possesses abundant hydropower resources. With a capacity of 34.61 million kilowatts, the latent hydropower potential of the basin's waterways is significant. This capacity is comprised of 29.32 million kilowatts, which represents 85 percent of the complete river system's potential. This region's abundant water resources and substantial hydropower potential make it a promising area for further academic exploration and hydropower development.

3.2. Feature selection

This study collected and organized monthly runoff data from the Lianghekou hydrological stations for the period of 1958 to 2018, a dataset comprising 130 climate indices from China, and meteorological data from four weather stations located within the control section. The meteorological data from the four weather stations include average dew point temperature, average precipitation, average station pressure, average temperature, average visibility, and average wind speed, sourced from the National Oceanic and Atmospheric Administration (NOAA) of the United States. The dataset of 130 climate indices mainly encompasses 88 monthly atmospheric circulation indices, 26 monthly sea surface temperature indices, and 16 other monthly indices, sourced from the China National Climate Center. Based on the atmospheric-ocean-terrestrial dataset, this study used the P-XGBoost-SHAP framework for factor dimensionality reduction and interpretation analysis, and used the reconstructed factors as the input of the forecasting models.

3.3. Performance assessment

In this study, four evaluation indexes, nash-sutcliffe efficiency coefficient (*NSE*), root mean square error (*RMSE*), mean absolute percentage error (*MAPE*) and relative error of flood volume (*VRE*), were used to evaluate the prediction performance of the model. The calculation methods of evaluation indexes are as follows:

$$NSE = 1 - \frac{\sum_{i=1}^n (Q_i - \hat{Q})^2}{\sum_{i=1}^n (Q_i - \bar{Q})^2} \quad (13)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (Q_i - \hat{Q}_i)^2}{n}} \quad (14)$$

$$MAPE = 100\% / n * \sum_{i=1}^n |Q_i - \hat{Q}_i / Q_i| \quad (15)$$

$$VRE = |\sum_{i=1}^n (Q_i - \hat{Q}) / \sum_{i=1}^n Q_i| \quad (16)$$

where, Q_i is runoff observation value, $\hat{Q}$ is runoff forecast value, n is the length of data sequence, and $\bar{Q}$ is the average value of runoff.

In general, the *NSE* reflects the degree of fit between the forecasted and observed values of hydrological elements during the flood forecasting process. A higher *NSE* value indicates a stronger fit between the forecasted and observed values. The *RMSE* measures the deviation between the observed and predicted values of hydrological elements during the flood forecasting process. A smaller *RMSE* value indicates a smaller degree of deviation between the forecasted and observed values. The *MAPE* assesses the accuracy of hydrological element predictions by quantifying the percentage deviation between observed and predicted values during the flood forecasting process. The *VRE* quantifies the degree of deviation between the observed total amount of hydrological elements and the forecasted total amount during the flood forecasting process. These metrics provide valuable insights into the quality and accuracy of flood predictions, with *NSE* indicating the goodness of fit, *RMSE* measuring the prediction deviation, *MAPE* assessing prediction accuracy, and *VRE* capturing the total amount prediction deviation in the flood forecasting process.

3.4. Comparative experiments

(1) Random Forest.

To validate the applicability and reliability of the proposed P-XGBoost factor reduction method in this study, a set of strongly correlated factors at the Lianghekou hydrological stations is still selected using the Pearson correlation coefficient. Random Forest (RF) is employed to assess factor importance and implement dimensionality reduction. Subsequently, the reduced factors are applied to the forecasting within the same interval, and a comparative analysis is conducted to evaluate the dimensionality reduction performance compared to XGBoost. Random Forest is implemented using the TreeBagger function in MATLAB.

(2) Empirical Mode Decomposition

This study conducts a comparative analysis between the Error Correction Method based on Empirical Mode Decomposition and Autoregression (EMD-AR) and the error correction method proposed in this paper to validate the superiority of the error correction model in improving the accuracy of runoff forecasting. The basic concepts and principles of Empirical Mode Decomposition (EMD) have been elucidated in section 2.3 and will not be reiterated here.

(3) Series-parallel coupling correction

The runoff forecast error correction is normally carried out using the series-parallel coupling correction method. This study also uses this method to compare with the EEMD-AR method to further verify the reliability of the error correction method. The principle of AR-based series correction is described in 2.3. Parallel correction is to combine the prediction results of multiple models, and the specific calculation method refers to the research results of (Wu et al., 2015).

4. Results and discussion

4.1. Prediction factors dimension reduction

This study performed Pearson correlation coefficient analysis on

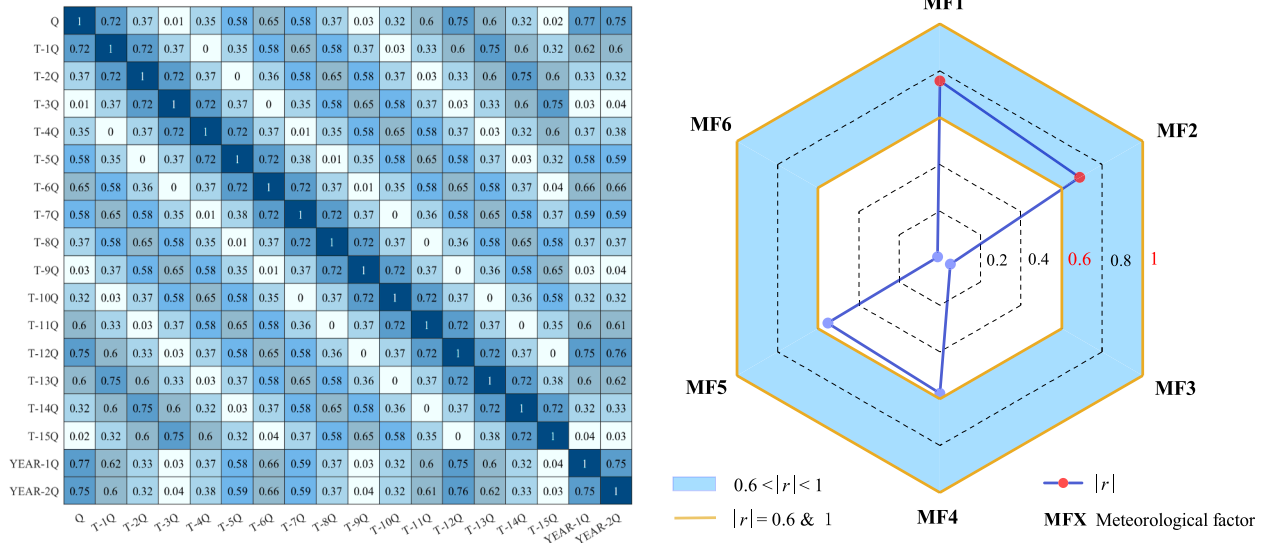


Fig. 5. Historical runoff, meteorological factor correlation coefficient heat map and radar map.

Table 2

The mean values of *NSE* and *RMSE* of the three models.

Model	Input dimension	Calibration <i>NSE</i> _{Avg}	<i>RMSE</i> _{Avg}	Validation <i>NSE</i> _{Avg}	<i>RMSE</i> _{Avg}
P-RF	5	0.8073	242.8073	0.7654	287.0528
	6	0.8102	241.1119	0.7615	289.5174
	7	0.8094	241.6731	0.7757	280.7295
	8	0.8177	235.5830	0.7647	287.8397
	9	0.8415	220.4826	0.7681	285.7610
	10	0.8399	221.4203	0.7745	282.3252
	11	0.8546	210.6853	0.7599	290.7891
	12	0.8513	213.6319	0.7607	290.2472
	13	0.8612	205.5335	0.7687	285.3854
	14	0.8579	208.2392	0.7676	285.9587
	15	0.8660	201.8434	0.7711	283.9124
	16	0.8693	199.1174	0.7684	285.6053
	17	0.8742	194.8542	0.7594	290.6211
	18	0.8729	195.9371	0.7549	293.0518
	19	0.8756	193.7057	0.7579	291.4388
	20	0.8742	194.2543	0.7542	293.3370
P-XGBoost	5	0.7888	254.5150	0.7834	276.0861
	6	0.8122	240.2754	0.8527	227.5548
	7	0.8268	230.5078	0.8537	226.6082
	8	0.8314	227.0950	0.8619	220.0117
	9	0.8492	214.3916	0.8668	216.0813
	10	0.8514	213.0526	0.8819	203.6050
	11	0.8505	213.4352	0.8690	214.4720
	12	0.8467	216.4396	0.8712	212.6704
	13	0.8481	215.5653	0.8710	212.6513
	14	0.8565	209.2255	0.8713	212.1476
	15	0.8655	202.7208	0.8705	212.8260
	16	0.8715	197.8327	0.8603	220.8144
	17	0.8743	195.0828	0.8625	218.9160
	18	0.8772	193.0486	0.8608	220.2203
	19	0.8844	186.7839	0.8563	223.8898
	20	0.8850	187.3559	0.8574	223.4335

monthly historical runoff data, meteorological factors and circulation index data of Lianghekou hydrological station from 1958 to 2018, and obtained the strong correlation factors set. Among them, the 15 historical average flow (T-1Q ~ T-15Q), 2 historical synchronous flow (YEAR-1Q, YEAR-2Q), 6 historical meteorological data (MF1 ~ MF6) and 130 circulation indexes (Atmospheric circulation indexes are expressed as AI1 ~ AI88; the sea temperature indexes are expressed as SI1 ~ SI26; other indexes are expressed as OI1 ~ OI16), a total of 153 dimensional factors, were used as the initial factors set. Then, the Pearson coefficient between the runoff and the initial factors set was calculated, and 51

strong correlation factors were obtained. The Pearson correlation coefficient of historical runoff, meteorological data is shown in Fig. 5.

To avoid the problems of complex model, data redundancy and slow training caused by too many input dimensions, it is necessary to find the best input dimension of the model. In this study, 51 strong correlation factor sets were used as feature inputs for XGBoost and RF, with 1961–2003 as the calibration period and 2004–2018 as the verification period. Finally, the importance evaluation value of each feature is obtained. In order to evaluate the actual performance of feature factors in the prediction model, this paper sorts them from high to low according

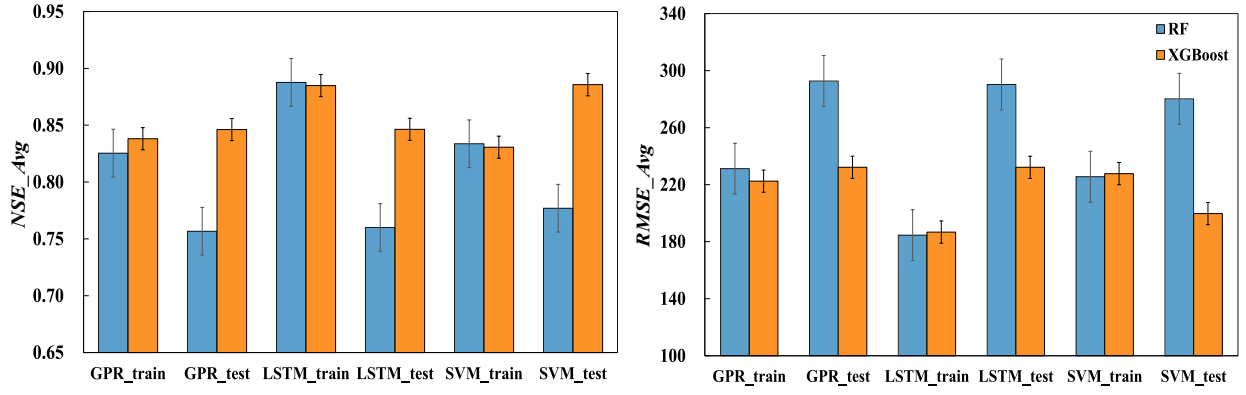


Fig. 6. The NSE_Avg and $RMSE_Avg$ of the models under the 5–20 input dimension.

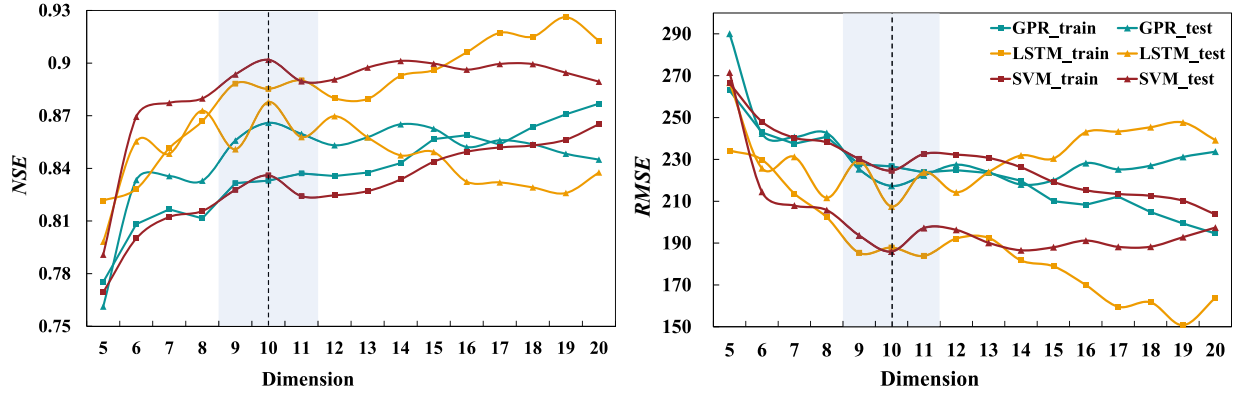


Fig. 7. The NSE and $RMSE$ under different input dimensions.

to the feature importance results of XGBoost and RF. Simultaneously, according to the ranking results, the feature factors of different dimensions are used as the input of GPR, LSTM and SVM models. Finally, NSE and $RMSE$ are used to evaluate the performance of the prediction model under different input dimensions.

Table 2 shows the mean values of NSE and $RMSE$ of the three models under the input dimensions of 5 ~ 20. As can be seen from Table 2, the calibration period evaluation indexes of P-RF and P-XGBoost are basically the same under different input dimensions, but the verification period indexes of P-XGBoost are significantly better than P-RF. This shows that the characteristic factors obtained by P-RF have more redundant information, which affects the generalization performance of the model. Fig. 6 shows the NSE_Avg and $RMSE_Avg$ of the forecasting models under the 5–20 input dimension. The performance of P-XGBoost in Fig. 6 is consistent with that of Table 2, which further verifies the reliability and superiority of P-XGBoost in feature dimensionality reduction.

Although the feature factors obtained by P-XGBoost are obviously better than that obtained by P-RF, with the increase of model input dimension, the NSE in calibration period continues to increase, and the NSE in verification period first increases and then decreases. This indicates an increasing risk of model overfitting. Therefore, it is necessary to select the appropriate input dimension to reduce the influence of redundant information on the prediction model. Fig. 7 visually shows the NSE and $RMSE$ under different input dimensions and prediction models, which can be used as the basis for determining model input dimensions.

It can be seen from Fig. 7 that when the input dimension of the model increases from 5 to 20, the NSE of each prediction model during the calibration period continues to increase, and the $RMSE$ also continues to decrease, but the NSE during the validation period shows a trend of

Table 3

Forecast factors of Lianghekou hydrological station.

Factor symbol	Represents meaning
T-1Q	1 month ago runoff
YEAR-2Q	The same period runoff two years ago
T-12Q	12 months ago runoff
T-6Q	6 months ago runoff
T-11Q	11 months ago runoff
MF2	Average precipitation (inches)
AI28	Eastern Pacific Subtropical High Ridge Position Index
MF1	Average dew point temperature
AI65	Tibet Plateau Region 1 Index
AI24	North African Subtropical High Ridge Position Index

increasing first and then decreasing. This is an over-fitting phenomenon caused by too many input dimensions of the model. This phenomenon is characterized by high accuracy of the model during the calibration period and low accuracy of the model during the validation period. Especially when the prediction model is LSTM, the over-fitting performance is more significant under high-dimensional input. Combined with Fig. 7, the input dimension of the model is more suitable in the range of 9 ~ 11. This study believes that the higher the NSE and the smaller the $RMSE$, the higher the generalization performance and reliability of the model. Finally, the input dimension of the model is determined to be 10, and the predictors after dimensionality reduction are shown in Table 3.

4.2. Prediction factors SHAP explanation

Based on the SHAP interpretability framework, this section performs global interpretation and feature interaction analysis for the 10 feature factors in Table 3. Meanwhile, Fig. 8 shows the SHAP feature importance

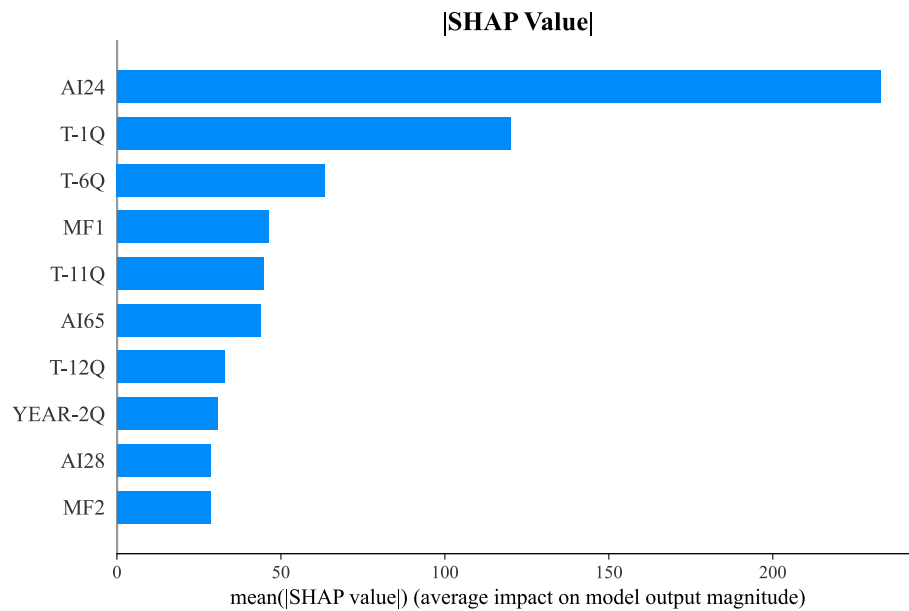


Fig. 8. The |SHAP Value| histogram of the top 10 features.

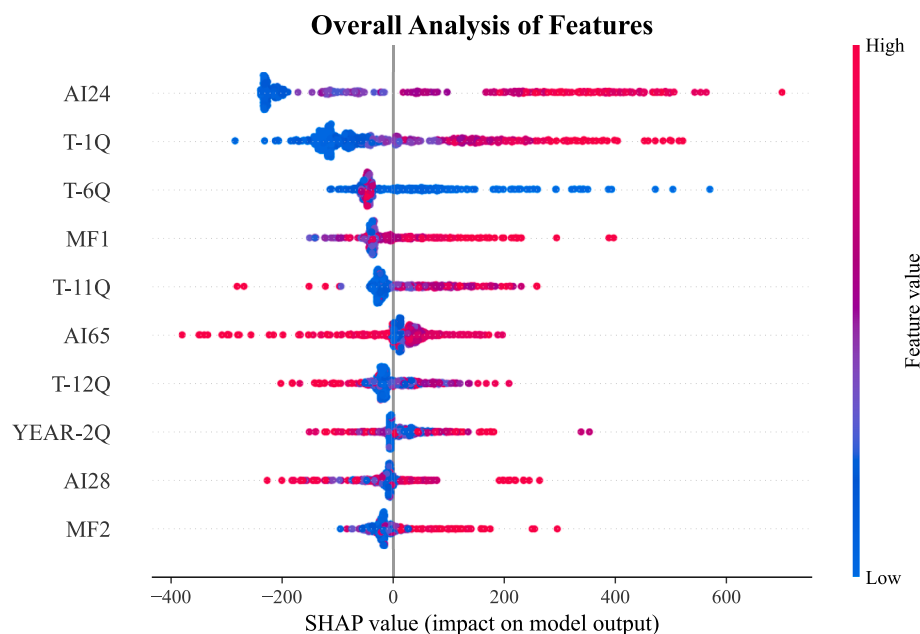


Fig. 9. Overall analysis of the top 10 features.

plots of the top 10 features. The mean value of the |SHAP Value| is the importance of each feature, which represents the degree of influence on the target output. It can be seen from the figure that the AI24 (North African Subtropical High Ridge Position Index) has the greatest contribution to the forecasting model, followed by T-1Q, T-6Q and MF1. The larger the SHAP value of these features, the greater the positive impact on the target output, and vice versa.

Fig. 9. Is the overall analysis of top 10 features in Table 3. This diagram not only intuitively presents the contribution of each feature and its positive and negative effects, but also expresses the relative strength of the eigenvalues through color. Red represents the high value, while blue represents the low value. Each row represents a feature, and the abscissa SHAP value represents the average influence of the feature. It can be seen from Fig. 9 that AI24, T-1Q and T-6Q have the greatest influence on the forecasting model. Among them, the larger the AI24 and

T-1Q values, the greater the positive impact on the output of the model. This shows that these two features play a core leading role in the model.

To further understand the interaction effect of features, Fig. 10 (a-h) shows the dependence plots of two features. It shows the complex interaction between features and the marginal impact on the forecasting model. The abscissa in the Fig. 10 is the range of feature values, the left longitudinal axis is the SHAP value, and the right longitudinal axis represents the relative size of another feature value. It can reveal the simple linear, monotonous or more complex relationship between the target and the feature. Fig. 10a and b show that there is an approximate proportional relationship between AI24, T-1Q and target output. However, T-6Q, AI65 and MF1 showed more complex correlations with target output. Meanwhile, MF1 and MF2 showed a weak positive correlation, indicating that there may be a synergistic effect between the two in promoting the prediction value of the model. Similarly, the

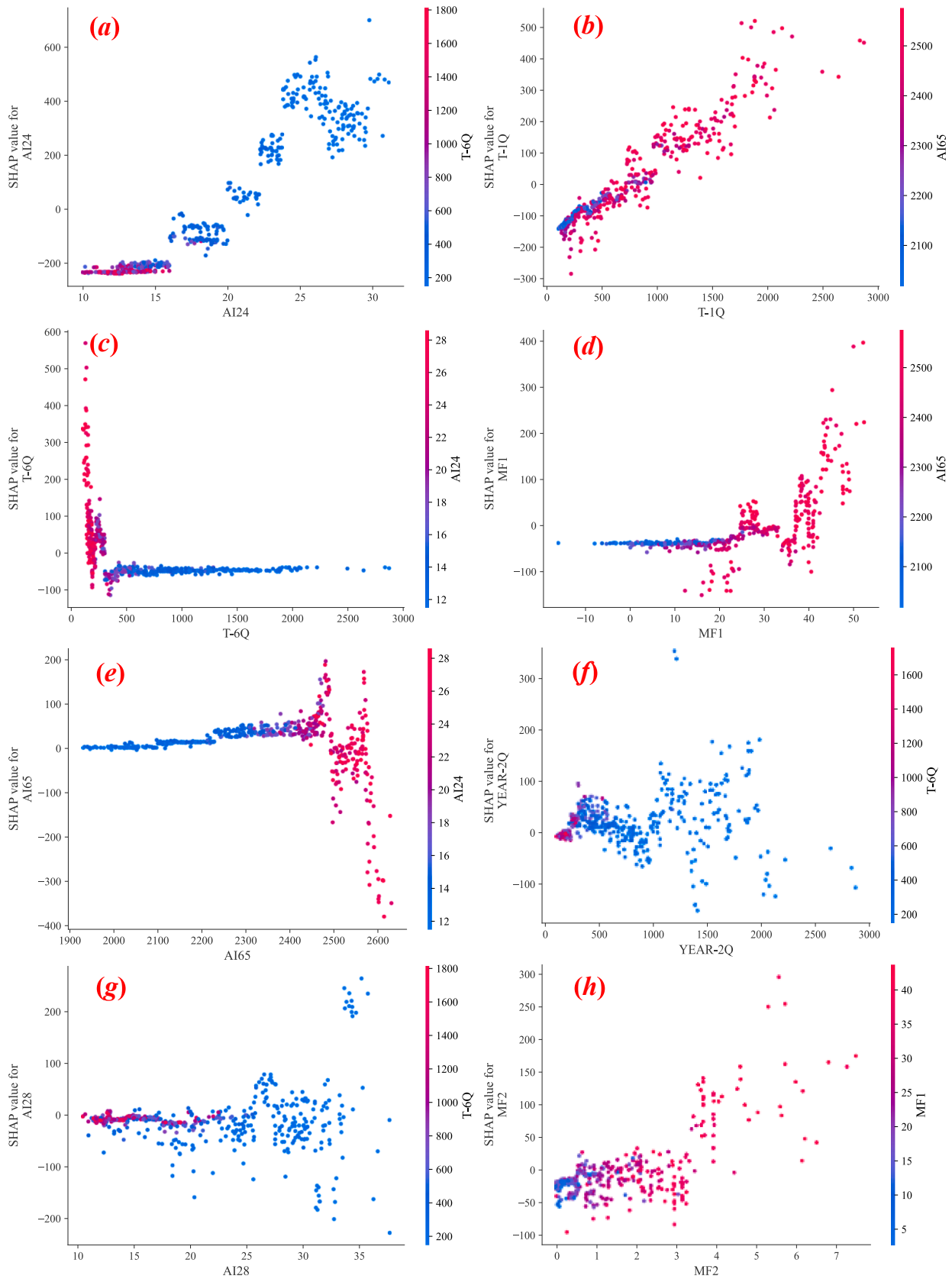


Fig. 10. The bivariate interaction scatter plots.

Table 4

Evaluation index of each model in Lianghekou hydrological station.

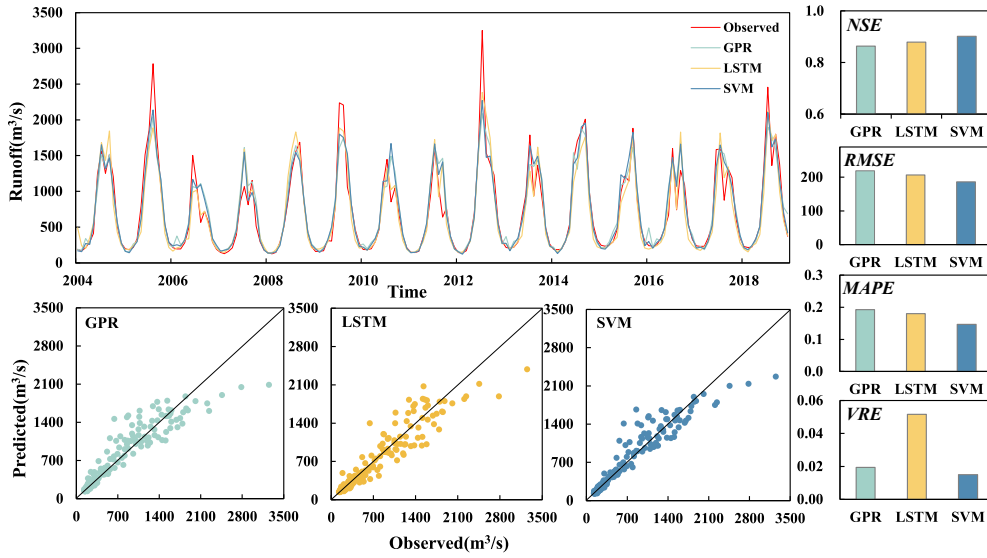
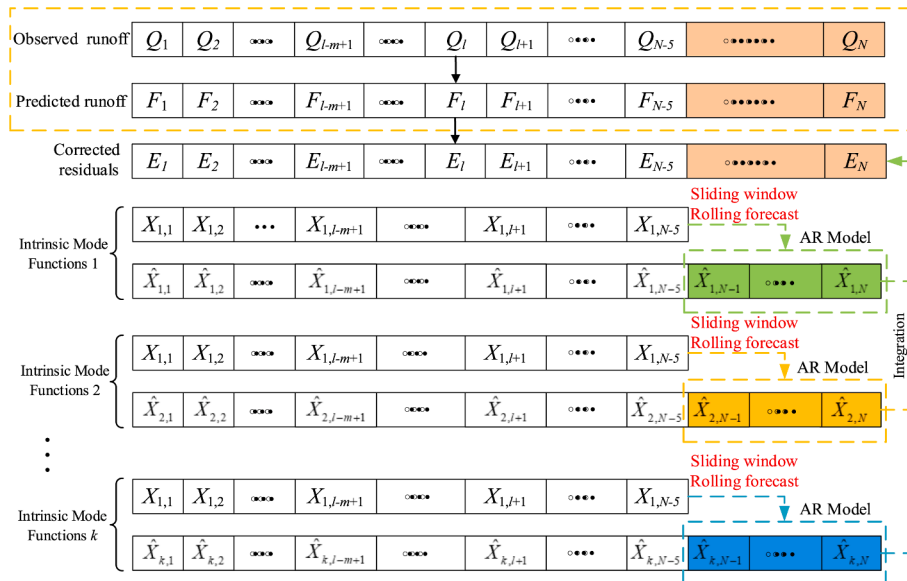
Method	Period	NSE	RMSE(m ³ /s)	MAPE	VRE
P-XGB-GPR	Calibration	0.8401	221.8275	18.27 %	0.0020
	Validation	0.8639	219.0395	19.25 %	0.0194
P-XGB-LSTM	Calibration	0.8821	190.4653	16.84 %	0.0008
	Validation	0.8788	206.6302	18.00 %	0.0516
P-XGB-SVM	Calibration	0.8360	224.6407	17.03 %	0.0238
	Validation	0.9019	185.9592	14.69 %	0.0150

features in other subgraphs show a variety of proportional or inverse relationships, reflecting the nonlinear relationship between features and target output to a certain extent. The above findings can deepen the understanding of the operation mechanism of the forecasting model and provide scientific support for the model accuracy improvement scheme.

4.3. Runoff forecasting model results analysis

Combined with the above feature factor dimensionality reduction and interpretation results, the final input dimension is determined to be 10. Based on this, this study used the data from 1961 to 2003 as the calibration period and the data from 2004 to 2018 as the verification period to verify the prediction effect of the three models. These three models are realized by MATLAB platform. The bold font in Table 4 shows that the index performs best in this period. Finally, four evaluation indexes of NSE, RMSE, MAPE and VRE corresponding to Lianghekou hydrological station were obtained, as shown in Table 4 and Fig. 11.

From Table 4, it can be seen that the NSE of each model in the calibration and validation periods is above 0.83, and the prediction accuracy in the verification period is higher, which can reach more than 0.86. Among them, the NSE during the validation period of the SVM model can even reach 0.9. Combined with the performance of the calibration period and the verification period, it can be judged that LSTM has the best overall performance and is more applicable in the basin, followed by SVM and GPR. The prediction results further verify the

**Fig. 11.** Forecast simulation diagram of each model validation period.**Fig. 12.** Five-step error correction flow chart.

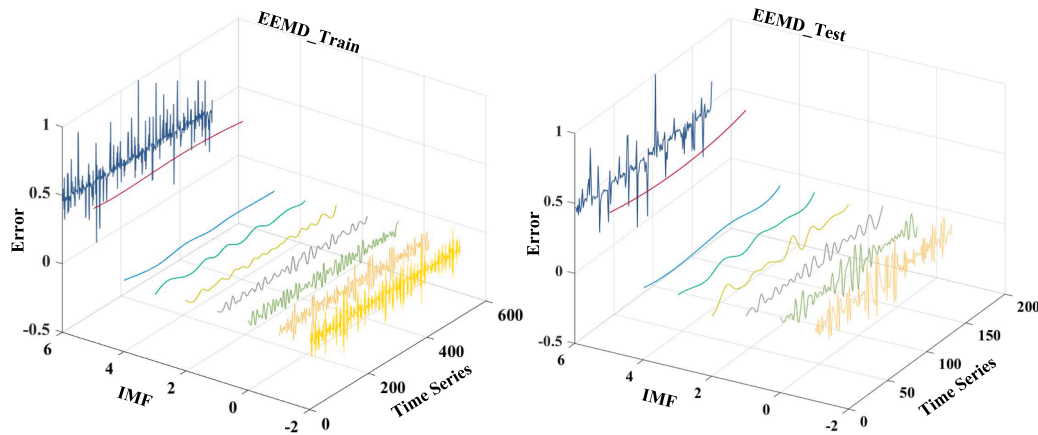


Fig. 13. EEMD decomposition diagram of SVM prediction residual error.

Table 5

Model evaluation index after error correction.

Method		Period	NSE	RMSE (m ³ /s)	MAPE	VRE
P-XGB-AR-Parallel		Calibration	0.8897	184.2522	18.93 %	0.0010
		Validation	0.9097	179.7694	19.80 %	0.0095
P-XGB-EMD-AR	GPR	Calibration	0.9129	163.6946	21.59 %	0.0015
		Validation	0.9102	179.3287	20.45 %	0.0227
	LSTM	Calibration	0.9146	162.1017	22.92 %	0.0055
		Validation	0.9119	177.5609	20.76 %	0.0106
	SVM	Calibration	0.9154	161.3682	22.06 %	0.0123
		Validation	0.9270	161.7048	19.38 %	0.0084
P-XGB-EEMD-AR	GPR	Calibration	0.9324	144.2592	19.36 %	0.0053
		Validation	0.9260	162.7381	20.68 %	0.0088
	LSTM	Calibration	0.9346	141.8625	19.94 %	0.0042
		Validation	0.9355	151.9424	17.18 %	0.0132
	SVM	Calibration	0.9335	143.0586	19.15 %	0.0050
		Validation	0.9380	148.9807	19.38 %	0.0004

superiority and reliability of the P-XGBoost-SHAP factor dimension reduction and interpretation framework in the construction of runoff prediction model.

4.4. Multi-step error correction results

In order to further improve the prediction accuracy of the model and explore the error propagation law, this study constructed a multi-step error correction framework based on EEMD decomposition strategy and AR method. However, EEMD and EMD decomposition strategies have the problem of data sequence information leakage. For solving this problem, the research work used the idea of sliding window and multi-step prediction to predict the IMFs. In the multi-step error correction process, it is not necessary to distinguish between model calibration and verification periods. However, in order to compare the improvement effect of the model before and after correction, this paper still divided the calibration period and verification period. The residual intrinsic mode functions are obtained by EMD and EEMD. In this study, the AR

method is used to predict the prediction residual sequence of the calibration period and the verification period in 5 steps. Finally, the runoff prediction value after error correction can be obtained. The multi-step error correction framework is shown in Fig. 12.

Based on the observed and predicted runoff values of Lianghekou hydrological station from 1961 to 2018, the residual sequence can be obtained by subtracting the two, and the residual sequence is decomposed by EMD and EEMD. Taking SVM as an example, the results of EEMD modal decomposition of the prediction residuals in the calibration period and the validation period are shown in Fig. 13. The residual term of modal decomposition has the largest amplitude, the trend is gentle, and it is easy to find its regularity. The low-frequency components IMF1 ~ IMF4 have good regularity and obvious periodicity, and the high-frequency components have obvious volatility. This part of the component is the part of the rapid change of the residual sequence.

This study employs EEMD to decompose-predict-integrate the residual sequence, and compare it with the traditional series-parallel coupling correction method (AR-Parallel) and EMD-AR decomposition method. The AR-Parallel method refers to the research of Wu et al (2015). Table 5 is the model evaluation index after error correction. It can be seen that the accuracy of runoff forecast is improved after error correction. Among them, the AR-Parallel correction method increases the runoff forecast NSE to 0.88 ~ 0.90 and reduces the RMSE to approximately 180 m³/s. The EMD-AR error correction method increases the runoff forecast NSE of each model to about 0.91, while reducing the RMSE to between 161 ~ 179 m³/s. The EEMD-AR error correction method increases the runoff forecast NSE of each model to about 0.93, while reducing the RMSE to a range between 141 ~ 179 m³/s. Compared with EMD-AR and AR-Parallel, the improvement effect of EEMD-AR method is the most significant.

Combined with Fig. 14, it can be seen that after EEMD-AR error correction, the prediction accuracy of each model is significantly improved. Meanwhile, the fitting degree of different models for flood peak is higher than that before error correction, which accurately reflects the trend of flood peak during the flood season. Fig. 15 shows the improvement of NSE and RMSE before and after error correction. It can be seen that the SVM model has the most best prediction effect after error correction, followed by LSTM and GPR.

Following error correction, the accuracy of the model validation period and the calibration period is improved to a certain extent, and compared with the EMD-AR error correction method, the EEMD-AR is more significant for the improvement of the prediction accuracy. From the perspective of each model, the SVM model has the most obvious improvement effect in the calibration period. Compared with the error correction, NSE has increased by 11.66 % while RMSE has decreased by 36.32 %. This is mainly due to the inaccuracy of the SVM calibration period prediction model. During the evaluation period of LSTM model,

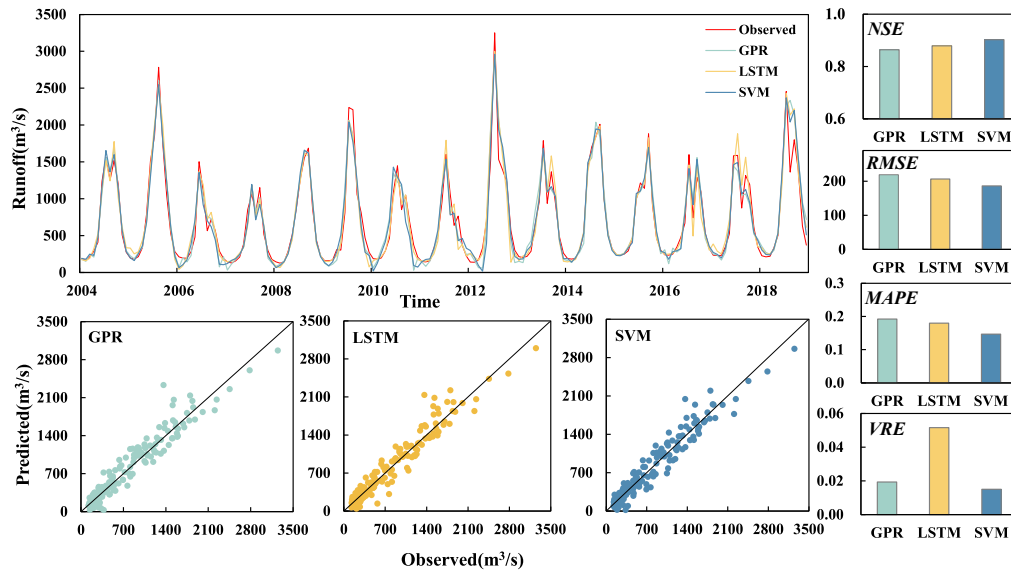


Fig. 14. The prediction simulation diagram of each model verification period after EEMD-AR.

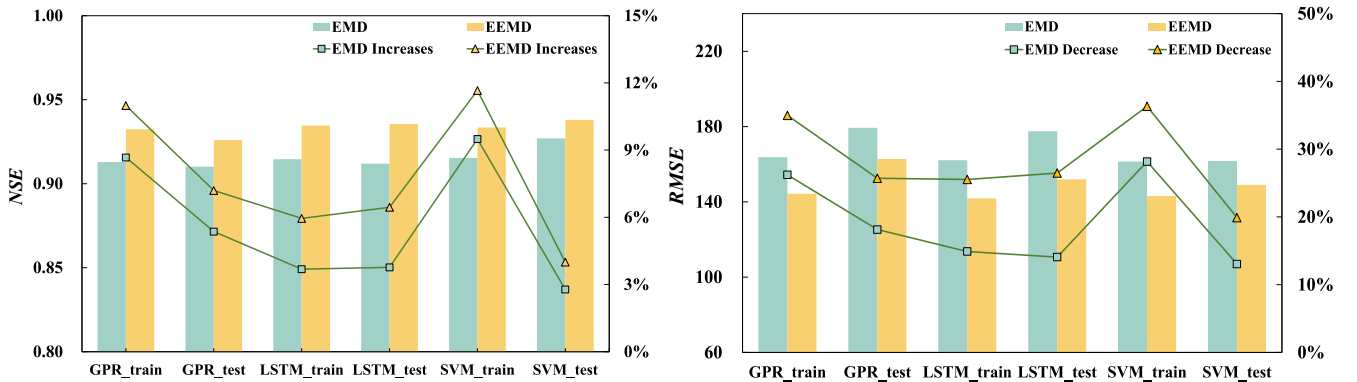


Fig. 15. NSE and RMSE before and after error correction.

Table 6

Model NSE and RMSE means with different rolling correction steps.

Step	Calibration NSE_Avg	RMSE_Avg	Validation NSE_Avg	RMSE_Avg
5	0.9323	144.3149	0.9347	152.6974
6	0.9307	146.0116	0.9403	146.1141
7	0.9305	146.1835	0.9402	146.2999
8	0.9291	147.6660	0.9396	146.9577
9	0.9264	150.5175	0.9375	149.5221
10	0.9276	149.1970	0.9358	151.4929
11	0.9248	152.0643	0.935	152.5159
12	0.9248	151.9818	0.9323	155.5986
13	0.9242	152.6211	0.9183	170.9201
14	0.9246	152.1918	0.9207	168.4416
15	0.9242	152.6769	0.9174	171.8768

NSE increased by 5.95 % and RMSE decreased by 25.52 % compared with that before error correction. The main reason is that the prediction accuracy of LSTM model is high, thereby limiting the enhancement effect.

In order to explore the influence of rolling correction step size on the error correction effect, this study sets the rolling correction step size range from 5 to 15. Meanwhile, the performance of the three prediction models (P-XGB-GPR, P-XGB-LSTM, and P-XGB-SVM) under different step sizes is deeply analyzed. Table 6 lists the mean values of NSE and RMSE of the three prediction models under different rolling correction

steps. Fig. 16 shows the trends of NSE and RMSE for the calibration and validation periods of the models with different rolling correction steps. The overall forecast accuracy of the model shows a decreasing trend as the rolling correction step increases. However, due to the differences in stability between models, some models show an increase in prediction accuracy when the step size is increased. This phenomenon reveals the complex effect of rolling correction step size on model performance.

Overall, the prediction factor dimension reduction method, runoff prediction model and error correction framework constructed in this study can better adapt to the characteristics of the Yalong river basin, improve the prediction accuracy and enhance the generalization ability of the model. However, after the runoff forecast results are corrected by EEMD-AR error, although the fitting degree of flood peak in flood season is obviously improved, the accuracy in dry season is reduced. This is mainly due to the fact that the waveform of the high-frequency modal component obtained by EEMD decomposition of the residual sequence changes sharply and the amplitude is small. The high-frequency component contains the fast-changing part of the residual sequence. These changes may be affected by noise or randomness, and the regularity is not obvious. At the same time, the sliding window and five-step rolling prediction can avoid information leakage, but with the increase of prediction step size, the prediction accuracy gradually decreases, which also affects the final prediction correction effect.

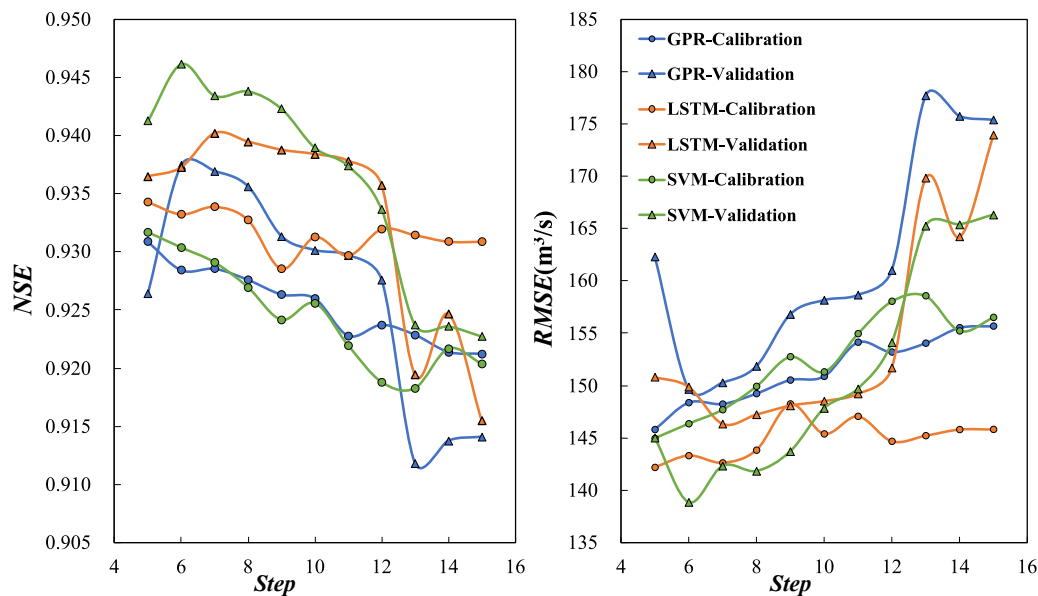


Fig. 16. NSE and RMSE values of the model under different rolling correction steps.

5. Conclusions

The study focuses on the Lianghekou hydrological station located in the Yalong river basin as the research object. Based on its historical runoff data, meteorological factors and circulation index dataset from 1958 to 2018, the factor dimensionality reduction and interpretation framework based on P-XGBoost-SHAP were proposed. Meanwhile, deep learning algorithms of GPR, LSTM and SVM were used to construct the runoff forecasting model. Finally, the multi-step error correction framework based on EEMD-AR is proposed for predicting residual errors. The results show that:

- (1) The P-XGBoost proposed in this study can not only evaluate the importance of factors, but also improve the prediction accuracy of the model. At the same time, SHAP explainable framework can deeply understand the global interpretation and interaction mechanism of runoff response factors, which is of great significance to improve the understanding of water cycle mechanism.
- (2) The prediction effect of each model in the dry season is better than that of the flood peak in the flood season, and the NSE in the validation period can reach more than 0.86. Overall, LSTM is more applicable in the Yalong river basin, followed by SVM and GPR.
- (3) For the residual error of runoff forecasting, the multi-step error correction method of EEMD-AR constructed by the research work has a stronger prediction error correction effect than the EMD-AR and AR-Parallel correction methods. It can increase the runoff forecast NSE to about 0.93, reduce the RMSE by more than 20 %, and significantly improve the runoff forecast accuracy.

This study still has the direction of further discussion, and further research and improvement can be made in the future. For example, the local value of the forecast result after error correction may not be as good as that of a single model. In the future, this study considers improving the modal decomposition method to make it suitable for runoff decomposition in natural state, so as to reduce the influence of high frequency component fluctuation. At the same time, the multi-model combination forecast method is discussed to further improve the forecast accuracy. Although this study only verified the runoff prediction effect in Lianghekou hydrological station, it can provide innovative solutions for reservoir inflow runoff and natural runoff prediction of other hydrological stations, and has great potential for application

value.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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